Event Driven CI/CD Pipelines

Enable Asynchronous CD without Skynet happening

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The Software Supply Chain

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The pipeline produces…

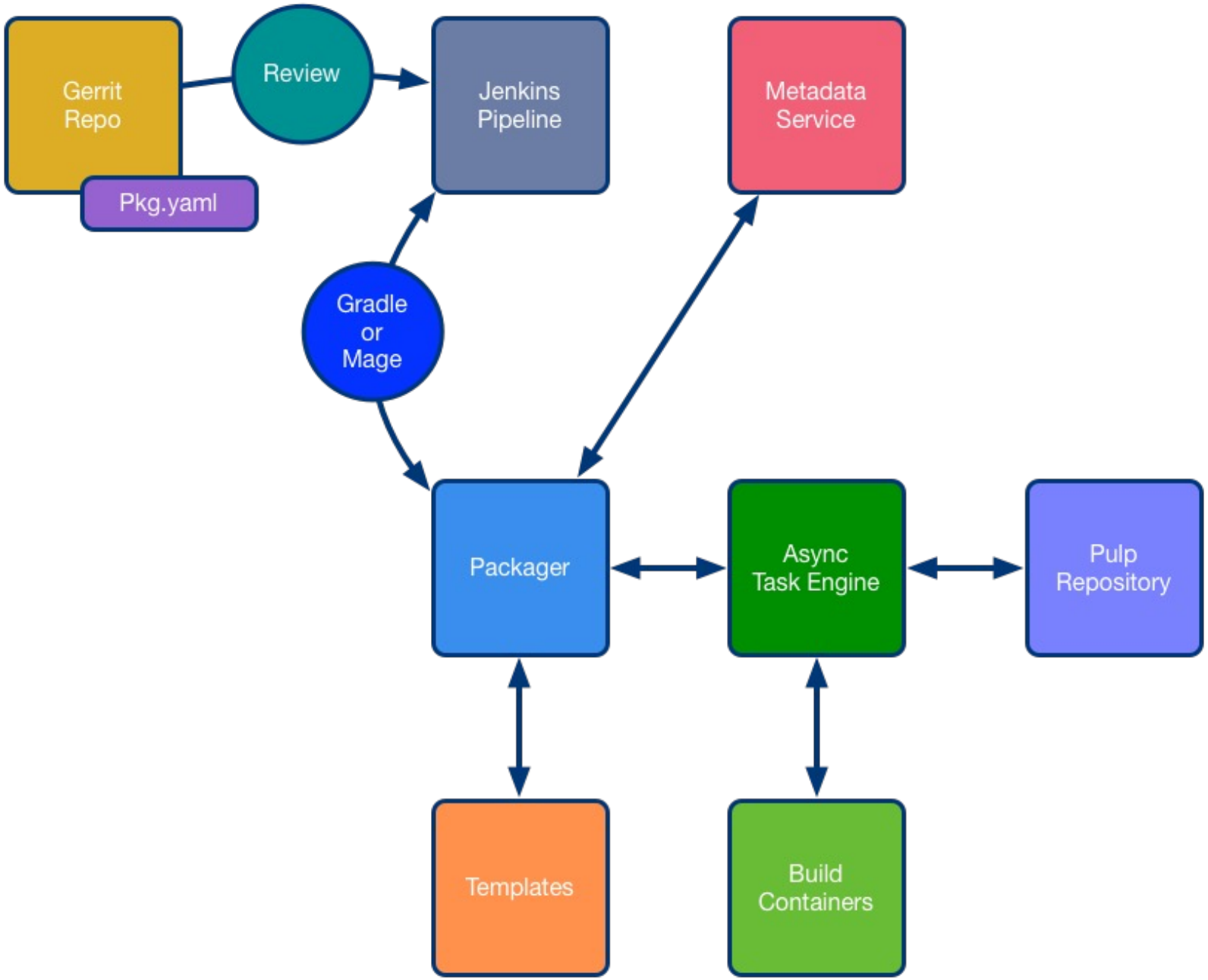
Packages
- RPMS
- DPKG
- OCI Images

Architectures
- x86_64
- aarch64
- ppc64le

Deployments
- Bare Metal
- VMs
- Kubernetes

*msi and msm packages as well

The pipeline before CI/CD Events



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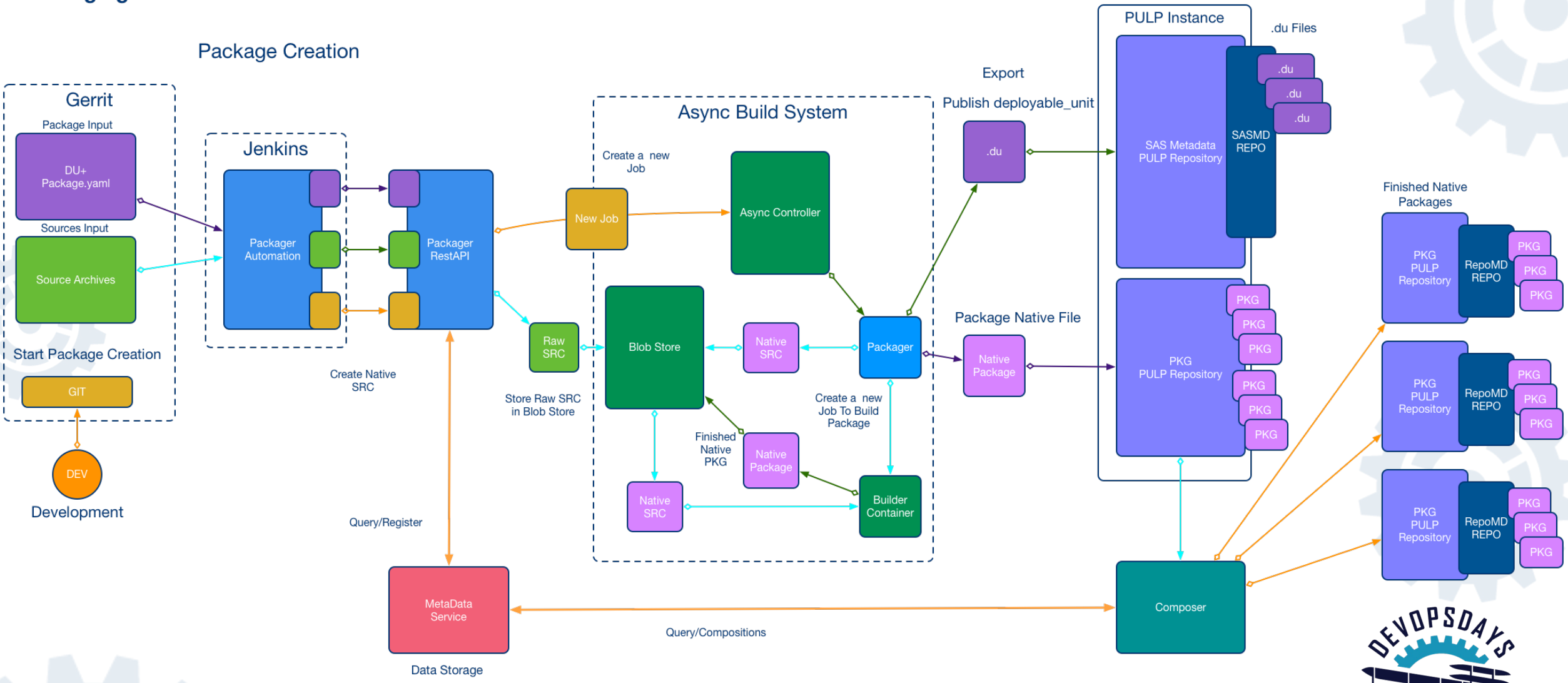
Packaging Automation



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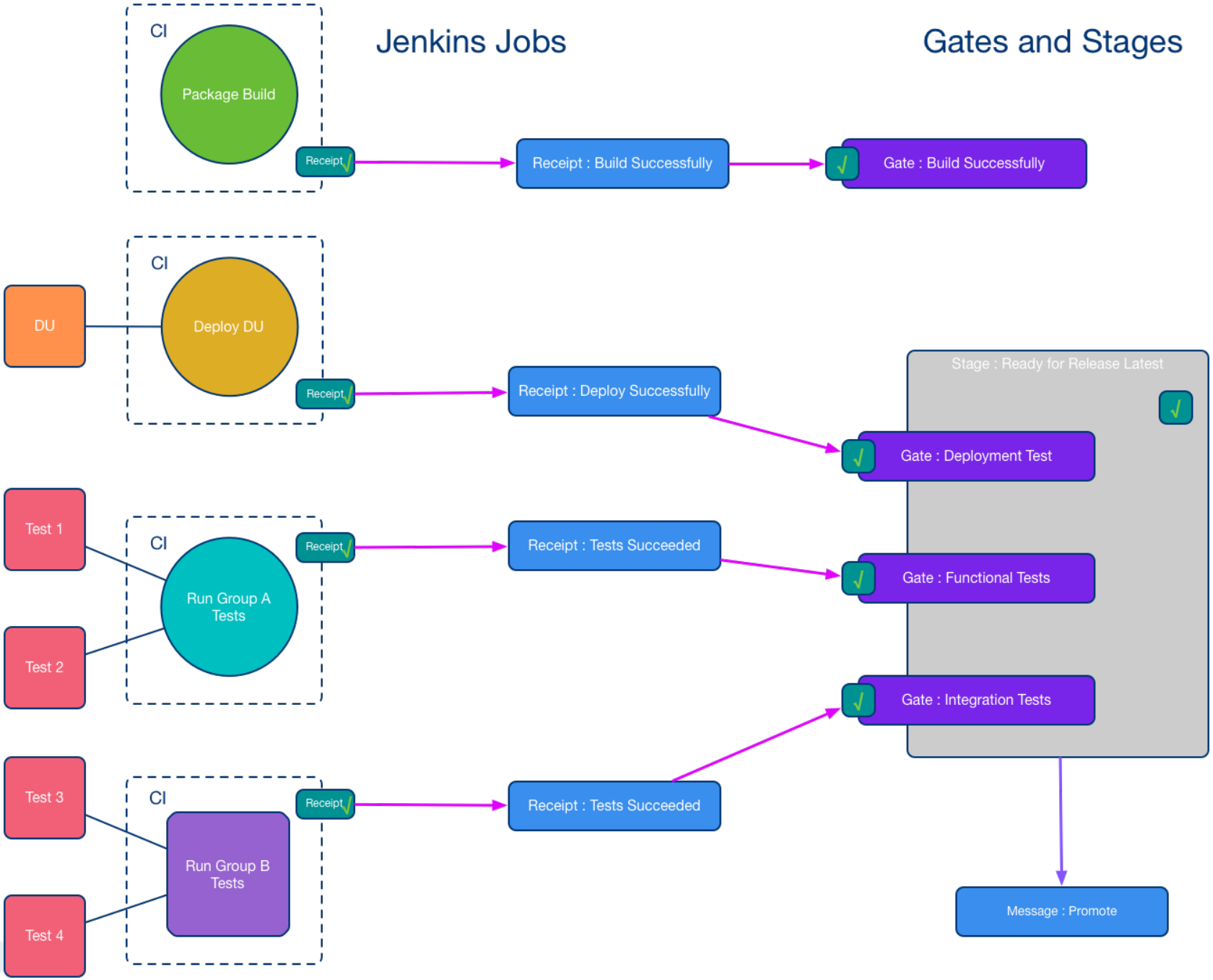
Why Event Driven CI/CD?

The Audit Trail and Gating

- Actions
- Receipts
- Gates
- Stages



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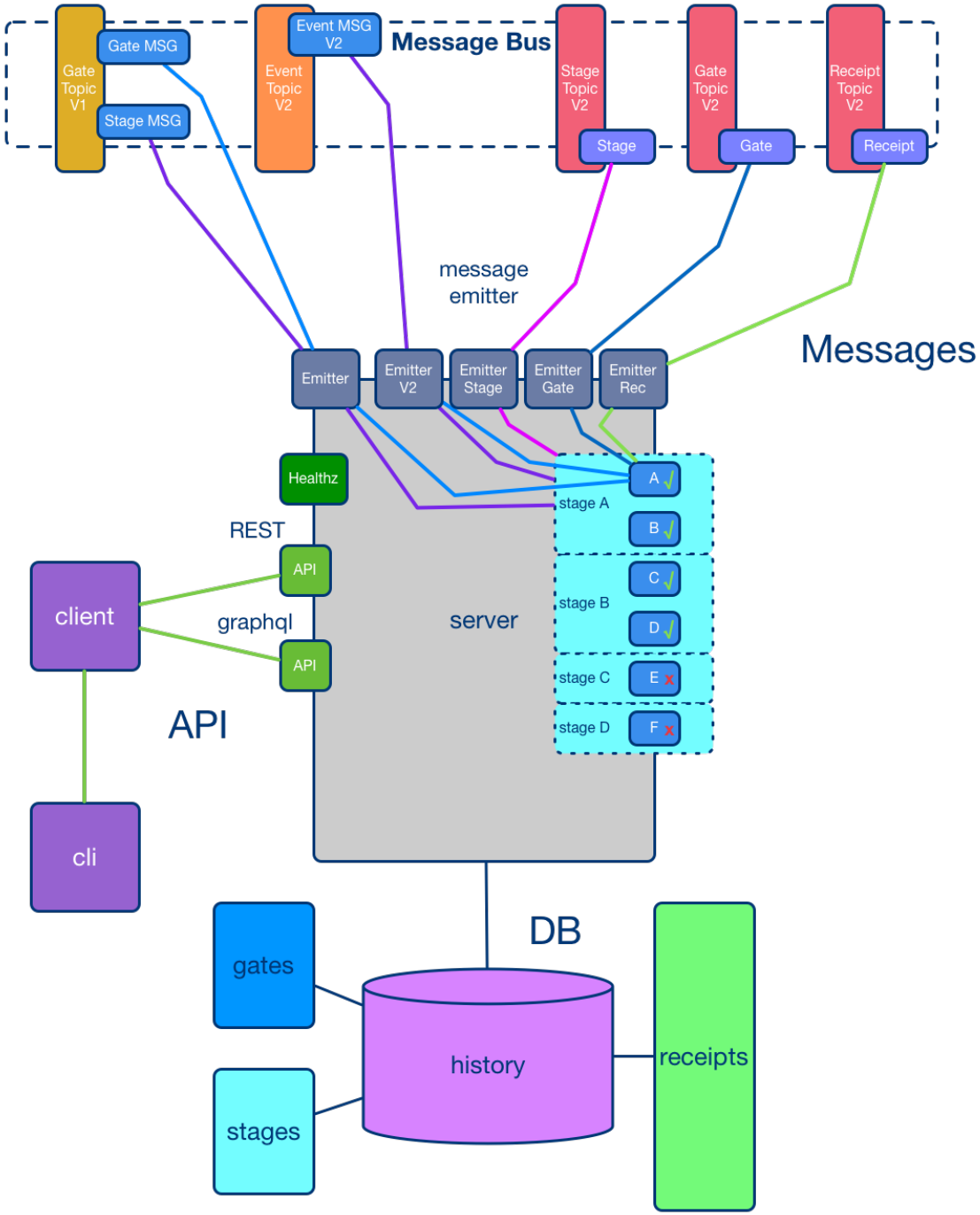

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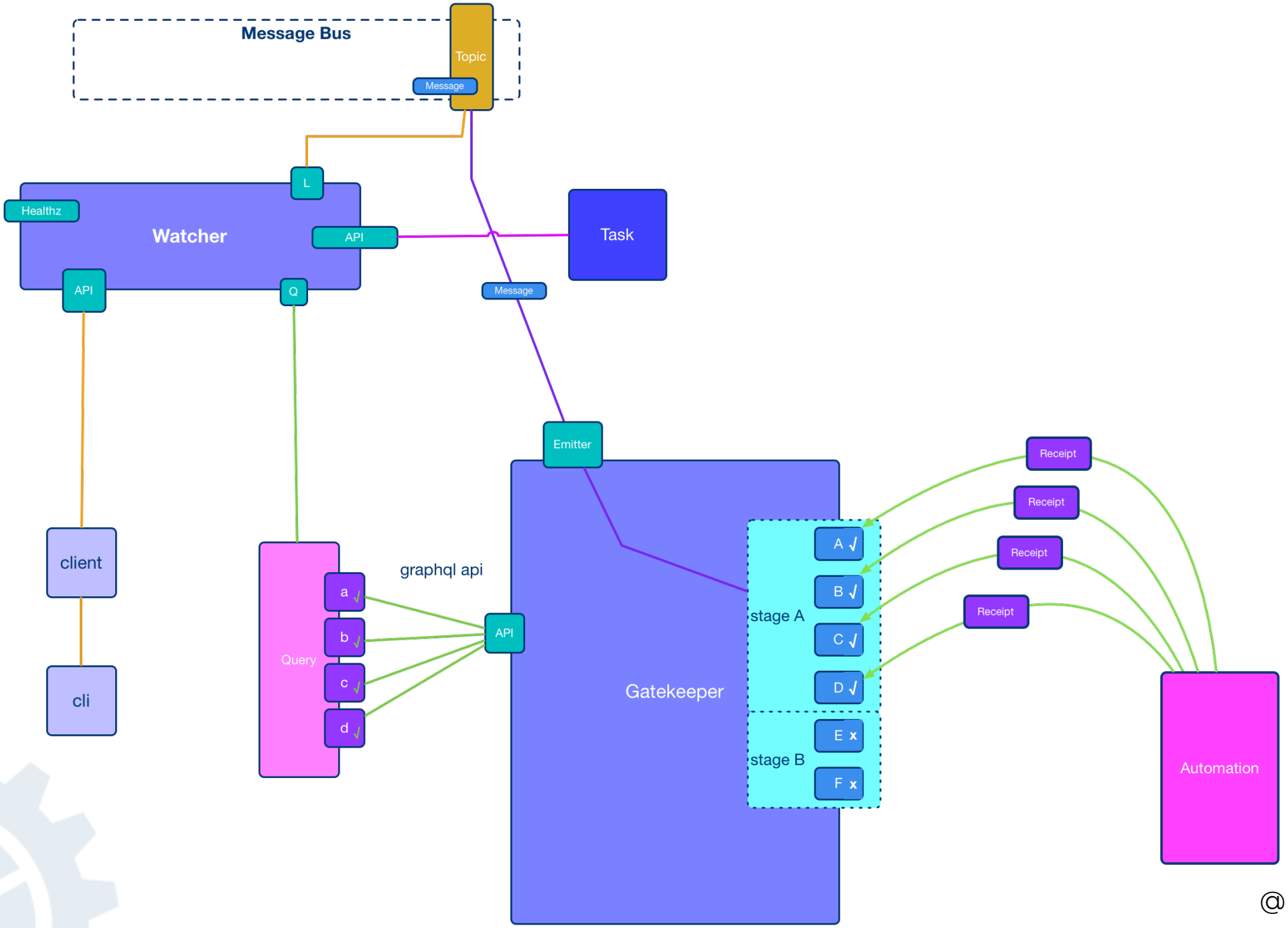

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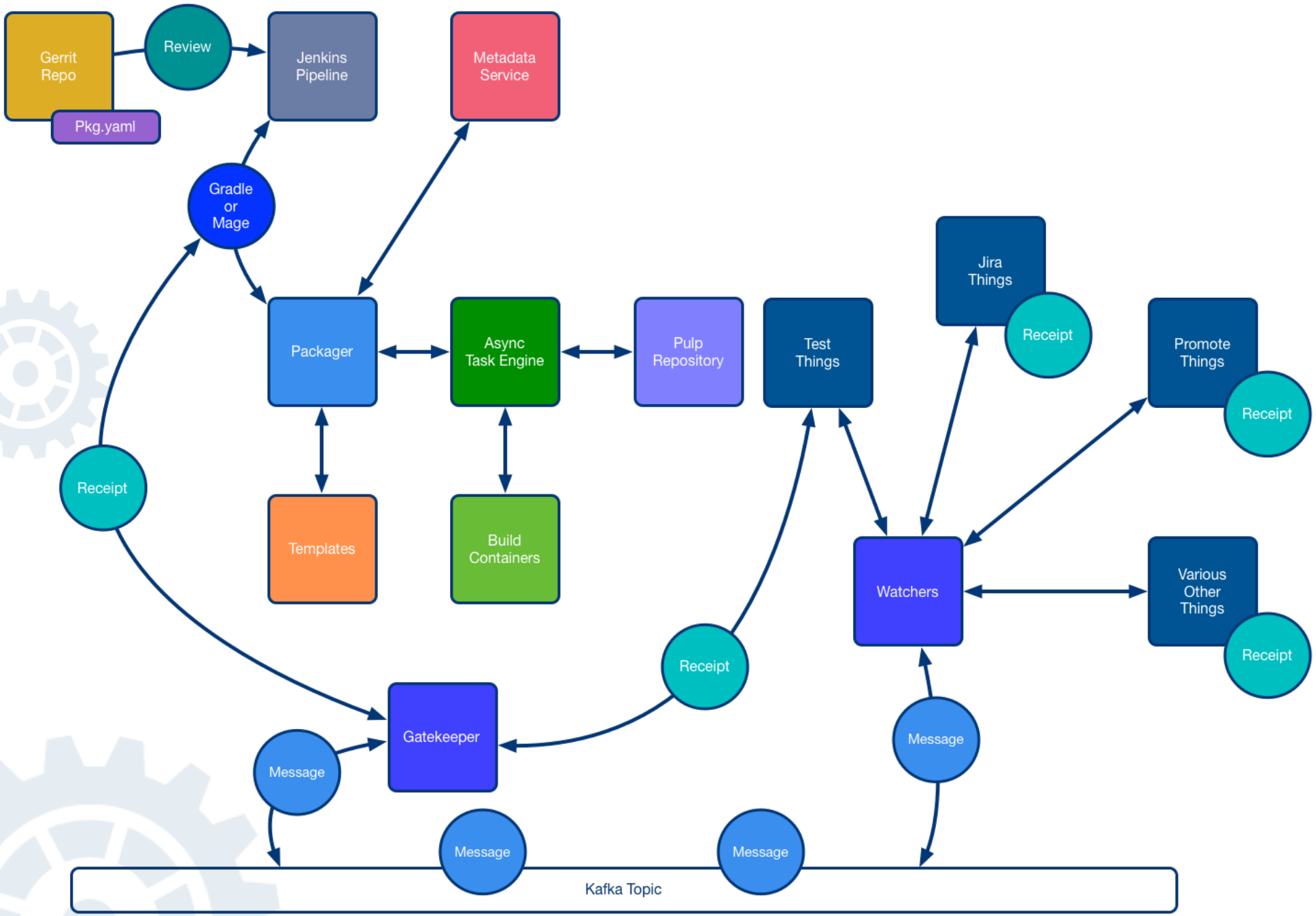
The pipeline after CI/CD Events



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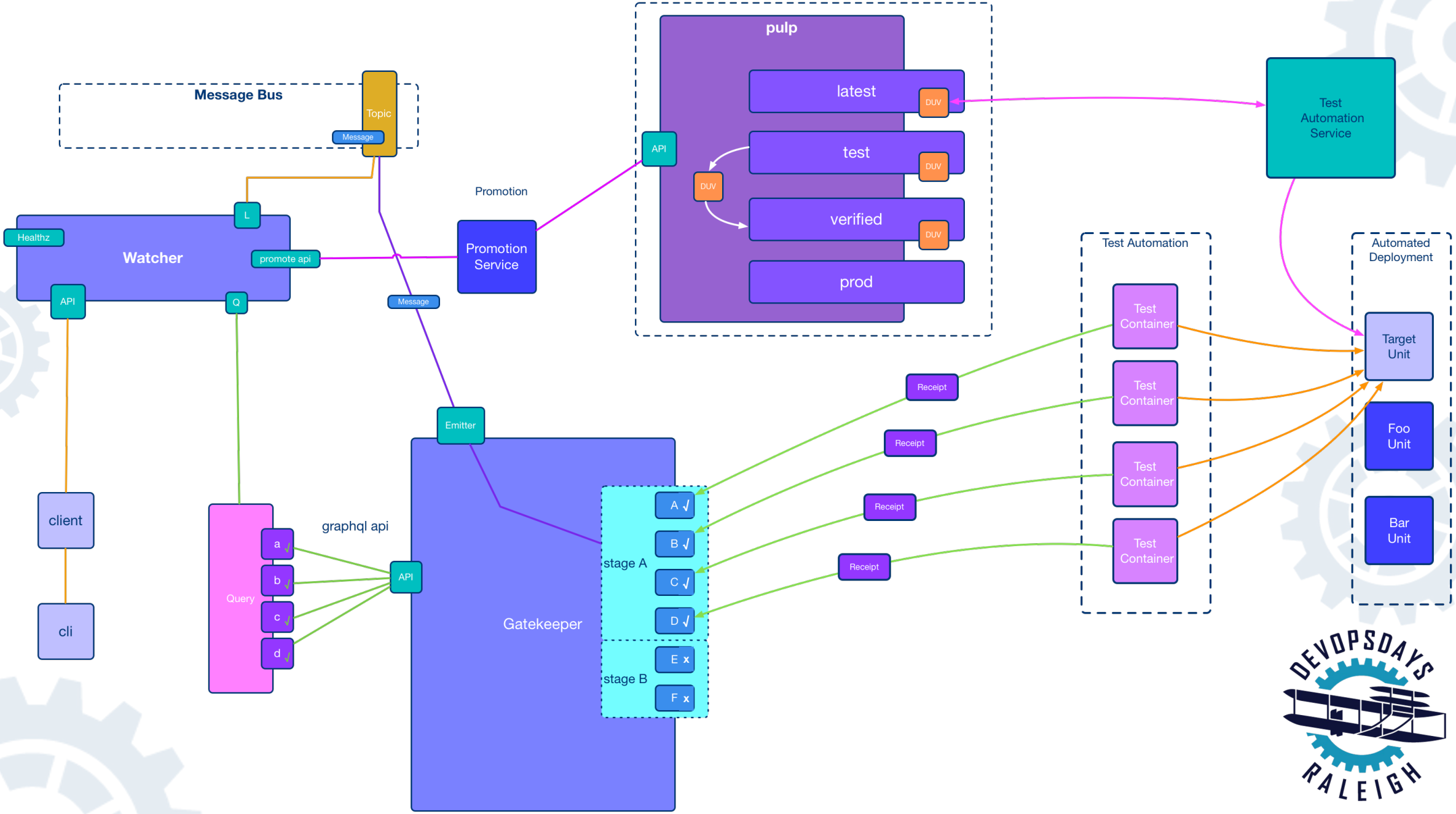

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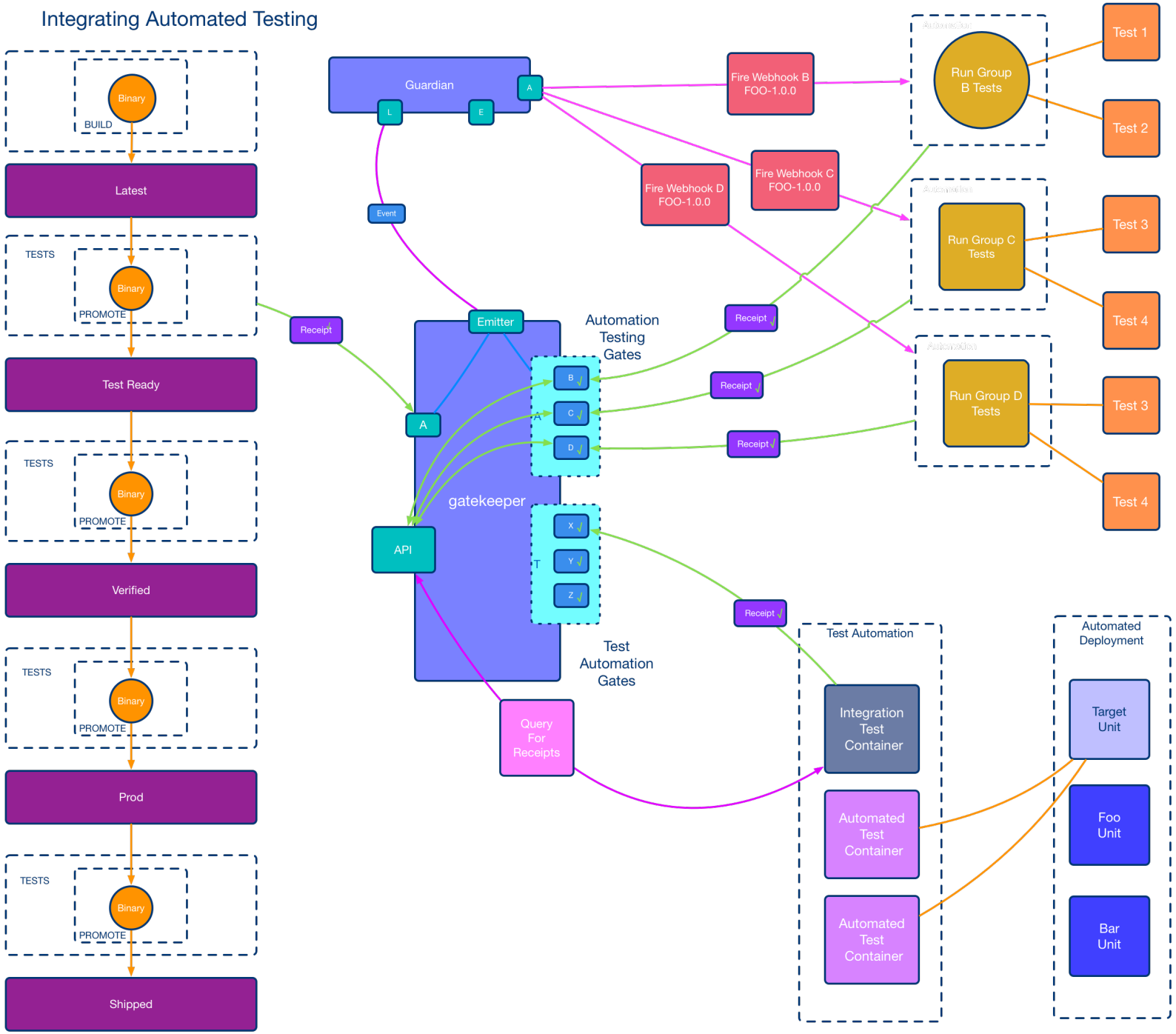
Integrating Automated Testing



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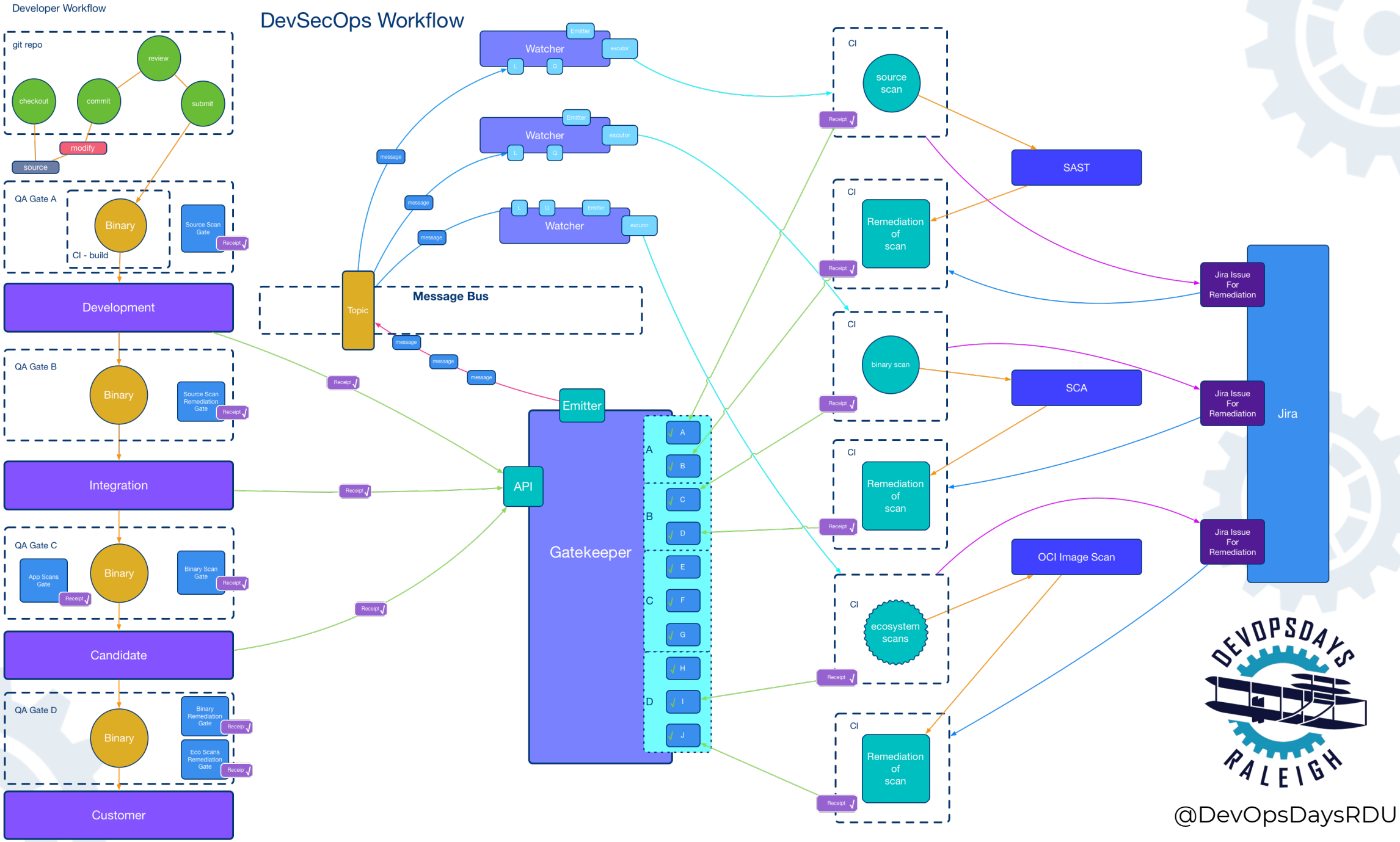
DevSecOps Workflow



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Lessons Learned

Observability

JSON Schema

Xunit Schema

Authentication (OIDC)

Guardrails

Watcher SDK

CNAB

Had we only…

Cryptographic Signing of Receipts

RBAC Service

Observability

Things that went well

Speed

Confidence

Quality

Security

Extendibility

CD Foundation

CDFoundation
https://cd.foundation

cdevents - A common specification for Continuous Delivery events
https://cdevents.dev

SIG Events
https://github.com/cdfoundation/sig-events

SIG Software Supply Chain
https://github.com/cdfoundation/sig-software-supply-chain

- Hope is not a strategy (truth)
- Electric Sheep harmed (0)
- Receipts, Gates, Stages (done)
- Everything gets a receipt (mostly done)
- Robots promote the units (done)
- Letter to Deckard (written)
- Automatic not just Automated (mostly done)
- Reproduce actions from Receipts (not quite how I wanted but close)
- Trace the lifecycle of a binary and how it got there (done)
- Machine learning to Optimize (not yet)
- Predictive test selection (not yet)
- Deployment Risk Prediction (not yet)
- Afraid of robots (true)

I am Smitty and I am afraid of Robots

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